INSTITUTIONAL-GRADE AV-OVER-IP ARCHITECTURE BLUEPRINT

Case Study: High-Density Redundant Dante Deployment for Premium Afro-Gospel & Afrobeats Full Band

1. Project Scenario & Input Profile

This blueprint maps out a high-capacity, ultra-low-latency digital audio network infrastructure optimized for a professional Nigerian live concert / mega-church stage setup. The environment requires uncompressed 24-bit/96kHz multi-channel audio distribution, glitch-free hardware redundancy, and a specialized personal monitoring ecosystem for the high-density rhythm section.

Input Distribution:

- **Vocals:** 1 Lead Vocal, 6 Background Vocals (BGV - Soprano, Alto, Tenor pairs)
- **Keyboards & Synths:** Master Keys (Stereo), Aux Synth (Stereo)
- **Guitars & Bass:** 3 Electric Guitars (Melody, Rhythm, Seiben), 1 Bass Guitar (Active Avalon U5 DI)
- **Drums & Percussion Suite:**
  - Acoustic Kit: Kick, Snare Top, Snare Bottom, Hi-Hat, Rack Tom, Floor Tom, Overhead Left, Overhead Right
  - Traditional Percussion: Gangan (Talking Drum), Akuba/Congas, Shaker, Woodblocks
- **Monitoring Subsystem:** High-density discrete In-Ear Monitors (IEMs) coupled with a dedicated daisy-chained Midas DP48 Personal Mixer network for the percussion team to allow localized, zero-latency dual-mix control over AES50/Dante bridging.

2. Audio-over-IP & Dante Signal Flow Map

3. Dante Controller Routing & Node Schedule

Transmitter Node	Ch	Signal Label	Bit Depth / Freq	Latency	Target Destination	Input Port	Application Assignment
Rio3224-Stage-01	01	Vocal_Lead	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 01	Lead Vocal Wireless
Rio3224-Stage-01	02-07	BGV_Sop_Ten	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 02-07	Core Background Vocals
Rio3224-Stage-01	08	Guitar_01_Mel	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 08	Electric Guitar 1 (Panned L)
Rio3224-Stage-01	09	Guitar_02_Rhy	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 09	Electric Guitar 2 (Panned R)
Rio3224-Stage-01	10	Guitar_03_Sei	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 10	Electric Guitar 3 (Seiben)
Rio3224-Stage-01	11	Bass_Avalon	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 11	Low-End Active Bass DI
Rio3224-Stage-01	12-19	DrumKit_Full	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 12-19	Main Acoustic Drums
Rio3224-Stage-01	20	Perc_Gangan	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 20	Talking Drum High Transient
Rio3224-Stage-01	21	Perc_Akuba	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 21	Traditional Conga / Akuba
Rio3224-Stage-01	22	Perc_Shaker	24-bit / 96kHz	0.25 ms	Console-SD7	Ch 22	Rhythm Section Shaker
Console-SD7	17-18	St_Perc_Stem	24-bit / 96kHz	0.50 ms	DN9650-Bridge	AES50 01-02	Downstream Percussion Group
Console-SD7	19-20	St_Drum_Stem	24-bit / 96kHz	0.50 ms	DN9650-Bridge	AES50 03-04	Downstream Drum Group Stem
Console-SD7	21-22	St_Guit_Stem	24-bit / 96kHz	0.50 ms	DN9650-Bridge	AES50 05-06	Panned Stereo Guitars Stem
Console-SD7	23	Mono_Bass_Stm	24-bit / 96kHz	0.50 ms	DN9650-Bridge	AES50 07	Low Frequency Sub-Pocket

Transmitter Node	Ch	Signal Label	Bit Depth / Freq	Latency	Target Destination	Input Port	Application Assignment
Console-SD7	24-25	St_Keys_Stem	24-bit / 96kHz	0.50 ms	DN9650-Bridge	AES50 08-09	Keyboards Stereo Bus
Console-SD7	26-27	St_Vox_Stem	24-bit / 96kHz	0.50 ms	DN9650-Bridge	AES50 10-11	Fully Blended Vocal Mix

4. Engineering Configuration Log & Architecture Rules

Midas DP48 & HUB4 Sub-System Execution:

- **Dual-Mix Optimization:** Because the Midas DP48 features dual 48-channel mixing engines, a single DP48 hardware surface on the percussion riser simultaneously runs two independent stereo IEM mixes. This allows the Gangan player (Mix A) and the Akuba player (Mix B) to sculpt separate spatial configurations while utilizing only one cable drop.
- **Reverb & Spatialization Modeling:** The DP48's onboard studio-grade reverb processor is locally engaged for the percussionists' earsets. This removes the processing load from the central front-of-house DiGiCo/Avid console and eliminates network latency that would occur if sending hardware aux effects return packets back down the line.
- **Local SD Card Capture:** Every DP48 station is configured to write uncompressed stereo WAV logs locally to onboard SD cards, ensuring multi-track archives of the rhythmic mixes can be instantly accessed post-event without consuming network recording bandwidth.

Network Protocol Rules:

- **IGMP Snooping:** Enabled globally across the main switch matrix. This guarantees that multi-channel multicast audio packages are strictly routed to the specified network interfaces and prevents the DP48/AES50 bridging architecture from being choked by unmanaged broadcast flooding.
- **QoS Egress Rules:** Strict Priority Queuing assigns DSCP 56 (CS7) explicitly to PTP synchronization traffic to enforce sub-millisecond clock accuracy across the network infrastructure.